

Week 1 Assignment Report

Objective

objective of this assignment is to build a strong foundation in Python programming for Data Analytics. Students will learn Python basics, data types, control flow, functions, collections, file handling, and apply these concepts by developing a mini Python project

Task 1: Python Basics

Create a Python program that: Prints a welcome message.

Takes your Name, College Name, and Branch as input.

Displays the entered information in a formatted output.

-----**code**-----

```
#Create a Python program that:
```

```
#Prints a welcome message.
```

```
#Takes your Name, College Name, and Branch as input.
```

```
#Displays the entered information in a formatted output.
```

```
# Task 1: Python Basics
```

```
# Print a welcome message
```

```
print("Welcome to the Python Basics Program!")
```

```
# Take input from the user
```

```
name = input("Enter your Name: ")
```

```
college = input("Enter your College Name: ")
```

```
branch = input("Enter your Branch: ")
```

```
# Display the entered information
```

```
print("\n----- Student Information -----")
```

```
print("Name   :", name)
```

```
print("College :", college)
```

```
print("Branch  :", branch)
```

```
print("-----")
```

-----**output**-----

```
rnova\week 1 tasks\task1.py'
Welcome to the Python Basics Program!
Enter your Name: Aahana
Enter your College Name: New LJ
Enter your Branch: CSE-AIML

----- Student Information -----
Name      : Aahana
College   : New LJ
Branch    : CSE-AIML
-----

(base) PS D:\internova\week 1 tasks>
```

Task 2: Variables & Data Types

Write a Python program that: Declares variables of different data types: Integer Float String Boolean Print each variable along with its data type using type()

-----code-----

```
# Task 2: Variables & Data Types
# Declaring variables of different data types
age = 21
percentage = 85.5
name = "Aahana"
is_student = True
# Printing each variable and its data type
print("Age:", age)
print("Data Type:", type(age))
print("Percentage:", percentage)
print("Data Type:", type(percentag))
print("Name:", name)
print("Data Type:", type(name))
print("Is Student:", is_student)
print("Data Type:", type(is_student))
```

-----output-----

```
Age: 21
Data Type: <class 'int'>
Percentage: 85.5
Data Type: <class 'float'>
Name: Aahana
Data Type: <class 'str'>
Is Student: True
Data Type: <class 'bool'>
(base) PS D:\internova\week 1 tasks> █
```

Task 3: Operators

Create a calculator program that performs: Addition Subtraction Multiplication Division Modulus using two numbers entered by the user.

-----**code**-----

```
# Task 3: Operators

# Take two numbers as input

num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

# Perform operations

print("\n---- Calculator Results ----")
print("Addition      :", num1 + num2)
print("Subtraction   :", num1 - num2)
print("Multiplication :", num1 * num2)
print("Division       :", num1 / num2)
print("Modulus        :", num1 % num2)
```

-----**output**-----

```
Enter first number: 10
Enter second number: 20

----- Calculator Results -----
Addition      : 30.0
Subtraction   : -10.0
Multiplication: 200.0
Division      : 0.5
Modulus       : 10.0
(base) PS D:\internova\week 1 tasks>
```

Task 4: Conditional Statements

Write a Python program that: Takes marks as input. Displays the grade using if, elif, and else.

Example: 90+ → A 75–89 → B 60–74 → C Below 60 → Fail

-----code-----

```
# Task 4: Conditional Statements
```

```
# Take marks as input
```

```
marks = float(input("Enter your marks: "))
```

```
# Display grade using if, elif, and else
```

```
if marks >= 90:
```

```
    grade = "A"
```

```
elif marks >= 75:
```

```
    grade = "B"
```

```
elif marks >= 60:
```

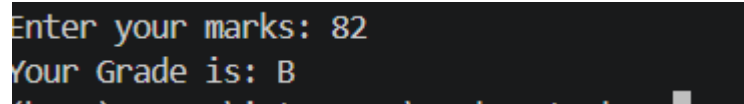
```
    grade = "C"
```

```
else:
```

```
    grade = "Fail"
```

```
print("Your Grade is:", grade)
```

-----output-----



```
Enter your marks: 82
Your Grade is: B
```

Task 5: Loops

Write Python programs to: Print numbers from 1 to 20 using a for loop. Print the multiplication table of any number. Print even numbers from 1 to 50 using a while loop.

-----**code**-----

```
# Print numbers from 1 to 20
```

```
for i in range(1, 21):
```

```
    print(i)
```

```
# Multiplication table
```

```
num = int(input("Enter a number: "))
```

```
for i in range(1, 11):
```

```
    print(num, "x", i, "=", num * i)
```

```
# Print even numbers from 1 to 50
```

```
i = 2
```

```
while i <= 50:
```

```
    print(i)
```

```
    i = i + 2
```

-----**output**-----

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

Enter a number: 4

4 x 1 = 4

4 x 2 = 8

4 x 3 = 12

4 x 4 = 16

4 x 5 = 20

4 x 6 = 24

4 x 7 = 28

4 x 8 = 32

4 x 9 = 36

4 x 10 = 40

2

```
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50
```

Task 6: Functions

Create two user-defined functions: Function to calculate the square of a number. Function to calculate the average of three numbers. Call both functions with user input.

-----code-----

```
# Task 6: Functions
```

```
# Function to calculate the square of a number
```

```
def square(num):
```

```
    return num * num
```

```
# Function to calculate the average of three numbers
```

```
def average(a, b, c):
```

```
    return (a + b + c) / 3
```

```
# Take input from the user
```

```

num = float(input("Enter a number to find its square: "))
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
c = float(input("Enter third number: "))

# Call both functions

print("Square of", num, "=", square(num))

print("Average of three numbers =", average(a, b, c))

```

-----output-----

```

Enter second number: 3
Enter third number: 5
Square of 9.0 = 81.0
Average of three numbers = 3.3333333333333335

```

Task 7: Strings & Collections

Create programs to demonstrate: String operations (upper(), lower(), replace(), find()) List operations (append(), remove(), sort()) Tuple creation and indexing Dictionary storing student information Set operations (add(), remove())

-----code-----

```

# Task 7: Strings & Collections

# 1. String Operations

print("----- String Operations -----")

text = "Python Programming"

print("Original String:", text)

print("Uppercase:", text.upper())

print("Lowercase:", text.lower())

print("Replace:", text.replace("Python", "Java"))

print("Position of 'Programming':", text.find("Programming"))

# 2. List Operations

print("\n----- List Operations -----")

numbers = [30, 10, 50, 20]

```



```
print("Original List:", numbers)
numbers.append(40)
print("After append:", numbers)
numbers.remove(10)
print("After remove:", numbers)
numbers.sort()
print("After sort:", numbers)

# 3. Tuple Creation and Indexing
print("\n----- Tuple Operations -----")
fruits = ("Apple", "Banana", "Mango", "Orange")
print("Tuple:", fruits)
print("First element:", fruits[0])
print("Third element:", fruits[2])

# 4. Dictionary storing student information
print("\n----- Student Dictionary -----")
student = {
    "Name": "Aahana",
    "Age": 21,
    "Branch": "CSE-AIML",
    "College": "LJ University"
}
print("Student Information:")
print("Name:", student["Name"])
print("Age:", student["Age"])
print("Branch:", student["Branch"])
print("College:", student["College"])

# 5. Set Operations
print("\n----- Set Operations -----")
```

```
subjects = {"Python", "Java", "C++"}  
print("Original Set:", subjects)  
subjects.add("SQL")  
print("After add:", subjects)  
subjects.remove("Java")  
print("After remove:", subjects)
```

-----output-----

```
----- String Operations -----  
Original String: Python Programming  
Uppercase: PYTHON PROGRAMMING  
Lowercase: python programming  
Replace: Java Programming  
Position of 'Programming': 7  
  
----- List Operations -----  
Original List: [30, 10, 50, 20]  
After append: [30, 10, 50, 20, 40]  
After remove: [30, 50, 20, 40]  
After sort: [20, 30, 40, 50]  
  
----- Tuple Operations -----  
Tuple: ('Apple', 'Banana', 'Mango', 'Orange')  
First element: Apple  
Third element: Mango  
  
----- Student Dictionary -----  
Student Information:  
Name: Aahana  
Age: 21  
Branch: CSE-AI ML  
College: LJ University  
  
----- Set Operations -----  
Original Set: {'Java', 'C++', 'Python'}  
After add: {'Java', 'SQL', 'C++', 'Python'}  
After remove: {'SQL', 'C++', 'Python'}
```

Task 8: Basic File Handling

Write a Python program that: Creates a text file. Writes your introduction into the file. Reads and displays the file contents.

-----**code**-----

```
# Task 8: Basic File Handling
```

```
# Create a text file and write introduction
```

```
file = open("introduction.txt", "w")
```

```
file.write("Hello! My name is Aahana.\n")
```

```
file.write("I am a Computer Science Engineering student.\n")
```

```
file.write("My branch is CSE-AIML.\n")
```

```
file.write("I am learning Python programming.")
```

```
file.close()
```

```
# Read and display the file contents
```

```
file = open("introduction.txt", "r")
```

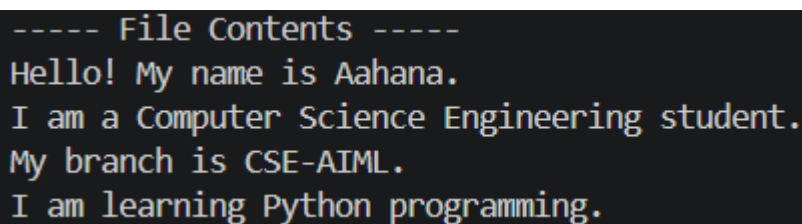
```
print("----- File Contents -----")
```

```
content = file.read()
```

```
print(content)
```

```
file.close()
```

-----**output**-----

A screenshot of a terminal window with a dark background. It displays the output of a Python program: a header line "----- File Contents -----" followed by four lines of text: "Hello! My name is Aahana.", "I am a Computer Science Engineering student.", "My branch is CSE-AIML.", and "I am learning Python programming." The text is in a light-colored monospace font.

```
----- File Contents -----  
Hello! My name is Aahana.  
I am a Computer Science Engineering student.  
My branch is CSE-AIML.  
I am learning Python programming.
```

Task 9: Mini Python Project

Develop a Student Record Management System using Python. The program should: Add student details. Display all student records. Search a student by name. Delete a student record. Use lists or dictionaries for data storage.

-----**code**-----

```
# Task 9: Student Record Management System
```

```
students = []
```

```
while True:

    print("\n===== Student Record Management System =====")
    print("1. Add Student")
    print("2. Display All Students")
    print("3. Search Student by Name")
    print("4. Delete Student")
    print("5. Exit")
    choice = input("Enter your choice: ")

    # 1. Add student details
    if choice == "1":
        name = input("Enter student name: ")
        age = input("Enter student age: ")
        branch = input("Enter student branch: ")

        student = {
            "Name": name,
            "Age": age,
            "Branch": branch
        }

        students.append(student)
        print("Student added successfully!")

    # 2. Display all student records
    elif choice == "2":
        if len(students) == 0:
            print("No student records found.")
        else:
            print("\n----- Student Records -----")

            for student in students:
                print("Name:", student["Name"])
```

```
print("Age:", student["Age"])
print("Branch:", student["Branch"])
print("-----")
```

3. Search student by name

```
elif choice == "3":
```

```
    search_name = input("Enter student name to search: ")
```

```
    found = False
```

```
    for student in students:
```

```
        if student["Name"].lower() == search_name.lower():
```

```
            print("\nStudent Found!")
```

```
            print("Name:", student["Name"])
```

```
            print("Age:", student["Age"])
```

```
            print("Branch:", student["Branch"])
```

```
            found = True
```

```
            break
```

```
    if not found:
```

```
        print("Student not found.")
```

4. Delete student record

```
elif choice == "4":
```

```
    delete_name = input("Enter student name to delete: ")
```

```
    found = False
```

```
    for student in students:
```

```
        if student["Name"].lower() == delete_name.lower():
```

```
            students.remove(student)
```

```
            print("Student record deleted successfully!")
```

```
            found = True
```

```
break
```

```
if not found:
```

```
    print("Student not found.")
```

```
# 5. Exit
```

```
elif choice == "5":
```

```
    print("Thank you for using the Student Record Management System!")
```

```
    break
```

```
else:
```

```
    print("Invalid choice. Please try again.")
```

-----output-----

```
1. Add Student
2. Display All Students
3. Search Student by Name
4. Delete Student
5. Exit
Enter your choice: 1
Enter student name: Aahana
Enter student age: 20
Enter student branch: cae-aiml
Student added successfully!

===== Student Record Management System =====
1. Add Student
2. Display All Students
3. Search Student by Name
4. Delete Student
5. Exit
Enter your choice: 2

----- Student Records -----
Name: Aahana
Age: 20
Branch: cae-aiml
-----

===== Student Record Management System =====
1. Add Student
2. Display All Students
3. Search Student by Name
4. Delete Student
5. Exit
Enter your choice: 5
Thank you for using the Student Record Management System!
```

❖ Code link

<https://drive.google.com/drive/folders/1AHJMJlyEeFqvtcud28DBCwxMZlIJonYI?usp=sharing>